

(a) What ProVerif proves, attacker by attacker

	breaks nothing	X25519 broken	ML-KEM broken	both broken
Full handshake				
Session data secret (also after both signing keys leak)	●	●	●	✗
Ground station knows it talks to the UAV	●	●	●	✗
UAV knows it talks to the ground station	●	●	●	●
1-RTT Cached RapidRekey				
New keys secret and agreed	●	●	●	●
... also if the next secret leaks afterwards	●	●	●	●
... if the old secret had leaked before	✗	✗	✗	✗
PQ ratchet				
New keys secret and agreed	●	●	●	●
... also if the old master secret leaks afterwards	●	●	●	✗
Old secret leaked before, attacker listens: heals	●	●	●	✗
Old secret leaked before, attacker active	✗			
Stored photo, recording, audio				
Content secret	●	●	●	✗
... also if the UAV is captured afterwards	●	●	●	✗
... also if the ML-KEM recording key is stolen	●	✗	●	✗
... also if the X25519 recording key is stolen	●	●	✗	✗
... if both recording keys are stolen	✗			
What is opened was sealed by the pinned UAV	●	●	●	●

(b) The libraries against NIST's vectors

Known-answer test	Pi	laptop
ML-KEM-1024 keyGen	25/25	25/25
ML-KEM-1024 encapsulation	25/25	25/25
ML-KEM-1024 decapsulation	10/10	10/10
ML-KEM-1024 encapsulation key check	10/10	10/10
ML-KEM-1024 decapsulation key check	10/10	10/10
ML-DSA-87 keyGen	25/25	25/25
ML-DSA-87 sigGen	30/30	30/30
ML-DSA-87 sigVer	15/15	15/15
random generator restored	2/2	2/2
X25519	2/2	2/2
HKDF-SHA-256	2/2	2/2
HMAC-SHA-256	2/2	2/2
SHA-256	1/1	1/1
AES-256-GCM	3/3	3/3

All 162 pass on both: yes.

liboqs 0.16.0: ML-KEM-1024 as "FIPS203",
ML-DSA-87 as "FIPS204"; OpenSSL 4.0.2.

(a): 46 ProVerif runs, 177 queries in 30 s. A dot is a claim that holds for any number of sessions; a cross is the same claim with the decisive scheme broken or key leaked, where the attack must be, and is, found.

No mark: not run, because the attack found with nothing broken exists all the more with something broken.

Symbolic model: ideal primitives, no timing, no sizes.

● proved ✗ attack found, as it must be